

Real Time Implementation of a Single Stage Converter based Solar-Wind Hybrid system

V.Rajini

Department of Electrical & Electronics Engineering
SSN College of Engineering
Chennai, India
rajini@ssn.edu.in

W. Margaret Amutha

Department of Electrical & Electronics Engineering
SSN College of Engineering
Chennai, India
margaretamutha@ssn.edu.in

Abstract- Increasing environmental pollutions, lack of power in remote places and demand for more energy makes us to seek new energy sources. Wind and solar hybrid energy have being popular ones owing to abundant, complement nature, ease of availability and convertibility to the electric energy. For hybridizing solar-wind system DC-AC or separate DC-DC converters are used one for each source. They will be connected to AD-DC or common DC-DC link to feed the loads according to the requirements. Such separate converters have the high count of elements and makes the system bulky and complex. This paper presents the performance comparison between separate cuk and buck converter and a single stage cuk-buck converter that integrates solar-wind renewable energy with battery backup using MATLAB/SIMULINK, to make best use of their operating characteristics and to achieve compactness and higher efficiency. FPGA controller is programmed for maintaining constant power at point of common coupling. An embedded controller based online state of charge estimation and battery charging system to suitably sink or source the input power based on the load demand is also carried out. Real time implementation of solar system with 1.5kW peak power, wind system with 1.4 kW and 48V, 200 Ah Lead-Acid batteries for powering 1200 Watts DC load is carried out to validate the theoretical approach.

Keywords---Hybrid solar-wind renewable energy system; MATLAB/SIMULINK; FPGA controller; State of charge estimation; Real time implementation

I. INTRODUCTION

The energy basically comes from fossil fuels, hydro sources, thermal sources and renewable sources [1-3]. With increasing threat of global warming and CO₂ emissions, many are going for renewable energy sources to preserve the earth for future generations. Renewable sources available are solar, wind, tidal, hydro, biomass and energy from waste [4]. The energy resources solar and wind are optional, at all times may not be available which causes an interruption in the power flow thus reducing the efficiency and consistency [5]. However due to unpredictable and intermittent nature of solar and wind systems, it has to be integrated with proper energy storage solutions. If these two energy sources are integrated, when a source is unavailable or insufficient in meeting the load demands, the other energy source can compensate the deficit [6-7].

In most renewable energy powered systems, input power delivered, output load demand, or instantaneous change in the both are not exactly identical with each other at any time instant. So providing a good match between them is a

complicated task. As reported by the authors [8-11], renewable energy sources were constructed using the topology which consists of multiple energy inputs connected to a respective DC-DC converter.

Each DC-DC converter is connected to a common dc link. The common dc link power is then distributed to the load. With this type of topology if one were to add multiple input modules, the overall system becomes too large and less efficient. According to the author [11-15], a single stage dual input converter can replace several numbers of parallel connected converters. It can also deliver power to the load either simultaneously or individually based on the battery utilization. This system is a better choice for the dual input hybrid systems and has many advantages such as low power components, simple structure and efficient.

An intelligent controller is much needed for this system because of non-linear VI characteristics and its dependency on sporadically varying natural phenomena [16-17]. It is needed to prioritize the sources for consumption, manage the power flow from sources to load, state of charge estimation, charging / discharging of batteries, instantaneous load demand and power management [18-19].

This paper is organized as follows. The first section is a short overview of the literature. In the second block diagram, system description and simulation results of separate and single stage cuk-buck converter are presented. Third section deals about the embedded controller and the results of state of discharge (SOC), charging and discharging of batteries. Fourth section contains the hardware description, results and efficiency calculation of the same. Finally, the findings of the investigations are highlighted in the conclusion.

II. BLOCK DIAGRAM AND SYSTEM DESCRIPTION

Fig.1 shows the block diagram of conventional and single stage converter used for green telecom. It contains two generating plants, a solar system and a wind system. These sources are connected in parallel to a -48V common dc bus thus feed DC load. The topology of conventional converter and cuk-buck converter powering DC load is given in Fig. 2 and Fig. 3 respectively. In Fig. 2 cuk converter is used to process power from solar panel, buck converter is used to process the power from wind system, both the outputs are connected to a common dc link which is used to power the load, therefore it is suitable for systems

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with low voltage rating like telecom. In Fig. 3, cuk-buck fused converter is used to process solar-wind hybrid system. It allows both simultaneous and individual operation of solar and wind sources.

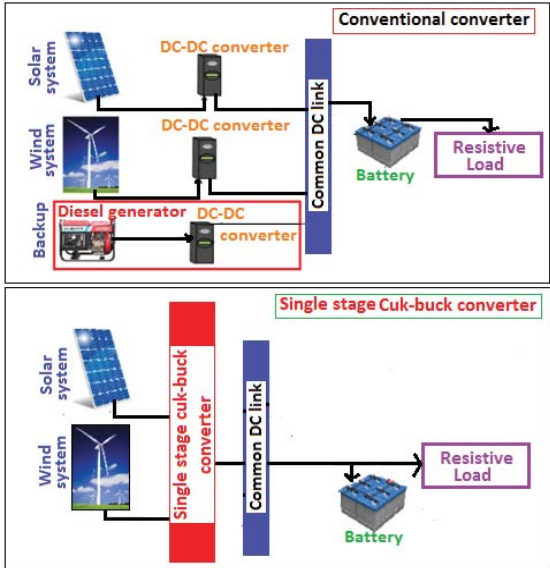


Fig. 1 Block diagram of conventional and single stage cuk-buck converter

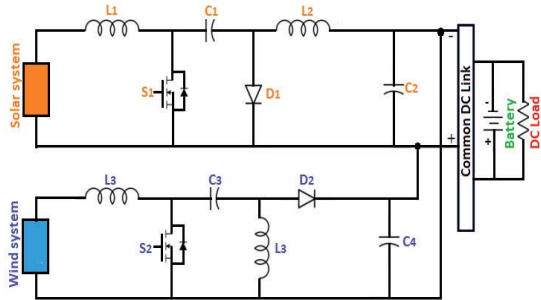


Fig. 2 Topology of conventional converter

Both the circuits are simulated with the parameters, L_1 of $158\text{ }\mu\text{H}$, L_2 of $84.16\text{ }\mu\text{H}$, C_1 of $2906\text{ }\mu\text{F}$, C_2 of $1132\text{ }\mu\text{F}$, resistive load of $1.92\text{ }\Omega$ and switching frequency of 20 kHz for its investigation. It is desired to obtain -48 V and 23 A from an inputs of 65 V and 55 V .

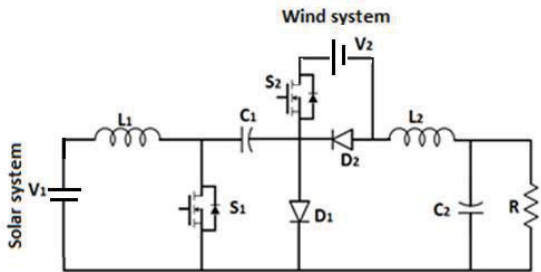


Fig. 3 Topology of single stage cuk-buck converter

A. Working principle of single stage converter

There are four different modes of operation depending upon the conduction states of the switches S_1 & S_2 as shown in Fig. 4.

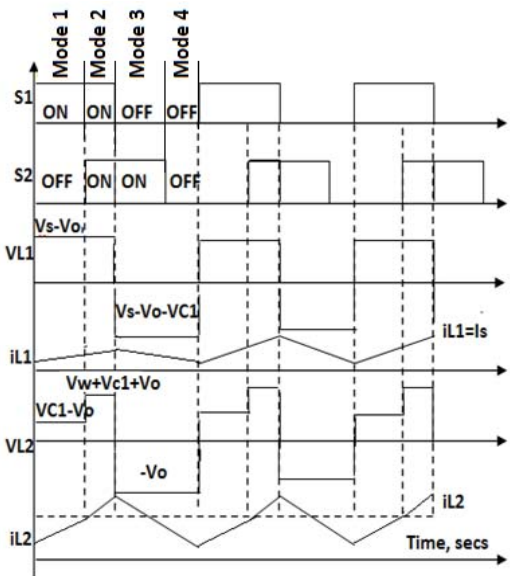


Fig. 4 Modes of operation
The working principle of different modes cuk-buck fused converter is explained in Table.1.

TABLE 1. WORKING OF CUK-BUCK CONVERTER

Mode	Switch	Working of conventional converter	Working of cuk-buck converter
Mode 1	S_1 -ON S_2 -OFF	The diode D_1 is in reverse biased. The power from solar charges the inductor L_1 , wind charges L_3 & L_4 . Energy stored in inductors L_2 & L_4 feeds the load and charges the battery.	The diode D_1 is in reverse biased and D_2 is in forward biased. The power from solar charges the inductor L_1 , energy stored in inductors L_2 & L_4 feeds the load and charges the battery and C_1 .
Mode 2	S_1 & S_2 ON	Diodes D_1 is in reverse biased. The power from solar charges the inductor L_1 . Energy stored in inductor C_2 & C_4 feeds the load as well as charges the battery.	Diodes D_1 , D_2 are in reverse biased. The power from solar charges the inductor L_1 . Energy stored in inductor C_3 feeds the load as well as charges the battery.
Mode 3	S_1 -OFF S_2 -ON	The diode D_1 is in forward biased. The energy stored in inductor L_1 & L_3 is transferred to capacitor C_2 . & C_4 which feeds the load and charges the battery.	The diode D_1 is in forward biased and D_2 is in reverse biased. The energy stored in inductor L_1 is transferred to capacitor C_1 . C_2 feed the load and charges the battery.
Mode 4	S_1 & S_2 OFF	The diode D_1 is in forward biased. Energy stored in L_1 is transferred to capacitor C_1 . Energy stored in L_2 & L_4 feeds the load and charges the battery.	The diodes D_1 , D_2 are forward biased. Energy stored in L_1 is transferred to capacitor C_1 . Energy stored in L_2 feeds the load and charges the battery.

B. Simulation results and discussion

The input voltages and gating pulses of the switches are shown in Fig. 5 and Fig. 6. The simulation results of conventional separate DC-DC converter are shown in Fig. 7. From the Fig. 7, it is seen that voltage across the inductor L_1 , L_2 , L_3 , L_4 are 48V, 50V, 40V and 40V. Capacitor voltages V_{C1} and V_{C3} are noted as 91.3V and 43V. Output voltage and current is measured across the capacitors C_2 and C_4 which is 46V and 23A. The average current through the inductors I_{L1} , I_{L2} , I_{L3} and I_{L4} are 22A, 21.5A, 22.365A and 15.08A respectively. The switch voltages S_1 and S_2 are measured as 60.01V and 50.15V. The efficiency of the converter is calculated as 71%.

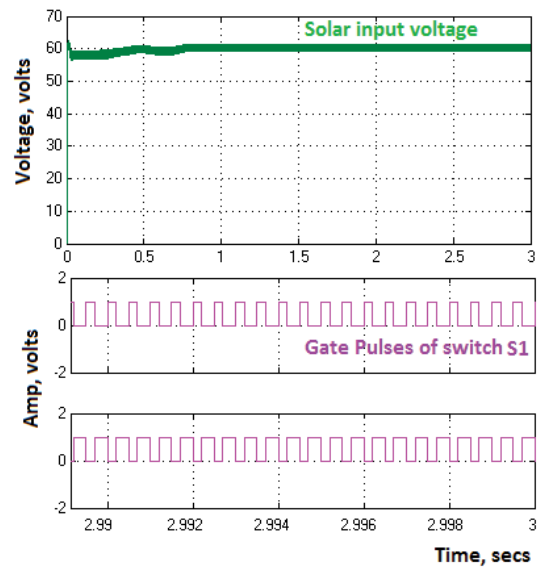


Fig. 5 Solar input voltage and switching pulse

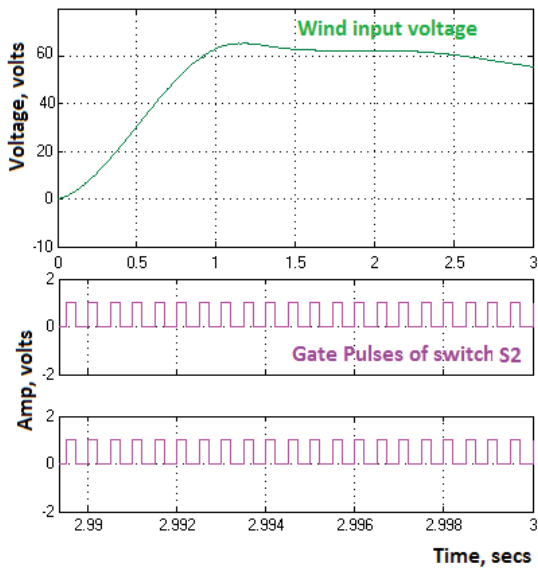


Fig. 6 Wind input voltage and switching pulse

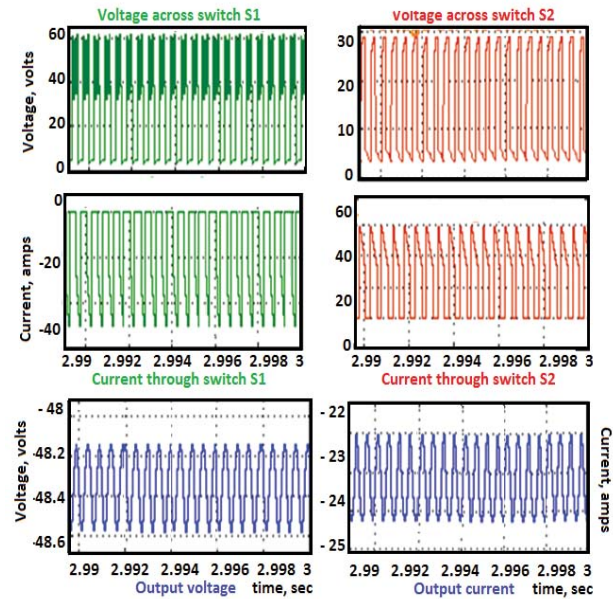


Fig. 7 Voltage across the switches, Output voltage and current of conventional system

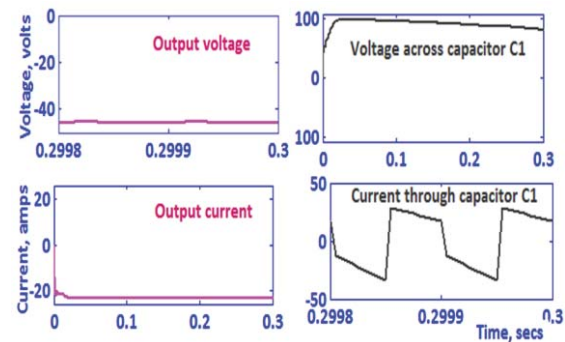


Fig. 8 Voltage and current waveforms of cuk-buck converter

Fig. 8 shows the output voltage and output current waveforms, voltage and current through the capacitor C_1 of cuk-buck converter. From the Fig. 8, it is clear that the output voltage and current obtained is -45.4V and -22.3A. The capacitor voltage measured is 88V. The average current of L_1 and L_2 is 22.9A and 22.4A respectively. The ripple factor of the inductors is calculated as 52.4 % and 53.5%. The energy storage in L_1 is 0.03J and copper loss is 6.29w. The energy storage in L_2 is 0.01J and copper loss is 6.02W. The overall efficiency of this type of converter is calculated as 89.16%.

III. EMBEDDED CONTROLLER BASED ENERGY MANAGEMENT SYSTEM

Embedded energy management system is the most important system in order to determine how effectively the power generated from renewable sources and distributed to the load. The controller decides when to use the sources or when to use the energy stored in the battery.

A. The control logic

The controller controls the converter by updating the operating current of the solar and wind sources via varying the duty cycle in a variable step manner till it satisfies the load demand.

Controller performances for the duty cycles d_s , d_w , d_{bc} , d_{bd} are given in Fig. 9. Thus the embedded controller for

energy management system is able to commend well for the different possible combinations of input powers to meet the load demand which is evidenced from the controller outputs d_s, d_w, d_{bc}, d_{bd} .

B. Embedded controller based SOC estimation of battery

The SOC is defined as the available capacity expressed as a percentage of its rated capacity. So, the battery is set at 40% state of charge (SOC) as depth of discharge (DoD) to about 70-80% of its capacity shall damage the battery. According to lead acid battery hand book the terminal voltage is an index of determining the SOC of the battery.

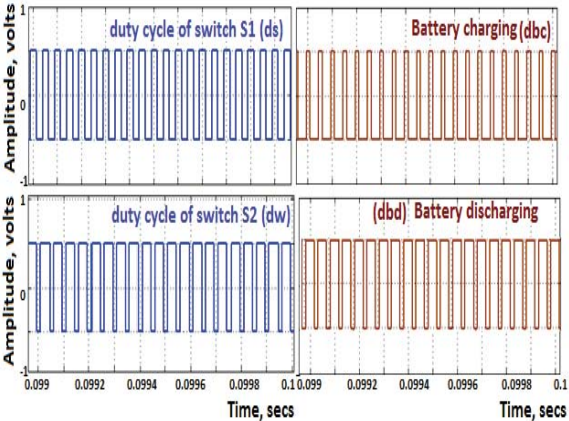


Fig 9 Controller performances

The SOC of 24V battery charging and discharging is estimated from terminal voltage and battery current. Table 2 and Table 3 show the terminal voltage at various charging currents and discharging currents for 24V lead acid battery. The charging and discharging process takes place depending on the current flow direction. The battery current direction and voltage are the inputs of the embedded controller based SOC estimation. The SOC estimation for the terminal voltage 24V and the current of 15A during charging process can be estimated by the following equation (1).

$$SOC = \left(\frac{(B-A) * (V-V_2)}{(V_2-V_1)} + C \right) \tag{1}$$

Where V= Battery voltage

C= Battery capacity

V_1 =Voltage lesser than the terminal voltage in the corresponding column of given current

V_2 =Voltage greater than the terminal voltage in the corresponding column of given current

A= V_1 row SOC reference

B= V_2 row SOC reference

TABLE 2 SOC VS TERMINAL VOLTAGE AT VARIOUS CHARGING CURRENT

SOC	At Charging			
	Voltage at (C/75)	Voltage at (C/50)	Voltage at (C/30)	Voltage at (C/15)
10	21	21.4	22	22.5
20	22	22.1	22.21	22.32
30	22.2	22.3	22.4	22.63
40	22.42	22.45	22.5	22.92
50	23.1	23.2	23.25	23.98
60	23.3	23.4	23.5	23.52
70	23.5	23.61	23.63	23.74

80	23.6	23.7	23.9	24
90	24.24	24.12	24.13	24.24
100	24.1	24.15	24.2	24.46

If the battery current or voltage lies between any two values specified in the Table 2 either in row or column wise, the SOC of battery for that voltage and current is determined by using equations (2), using the row wise neighbouring values of voltage between which the battery current falls. For the estimation of charging current 20A , the battery current lies in between the columns (C/15) and (C/30), then a new column should be created virtually and then SOC needs to be estimated using the following formula,

$$V_1 = \left(\frac{(22.5 - 22) * \left(I - \frac{C}{30} \right)}{\left(\frac{C}{15} - \frac{C}{30} \right)} + 22 \right) \tag{2}$$

$$V_2 = \left(\frac{(22.32 - 22.21) * \left(I - \frac{C}{30} \right)}{\left(\frac{C}{15} - \frac{C}{30} \right)} + 22.21 \right) \tag{3}$$

$$V_3 = \left(\frac{(22.63 - 22.4) * \left(I - \frac{C}{30} \right)}{\left(\frac{C}{15} - \frac{C}{30} \right)} + 22.4 \right) \tag{4}$$

Likewise V_4, V_5 till V_{10} is calculated. Then calculate SOC using the equation (1). Similarly SOC estimation for discharging process is calculated using the above procedure and discharge data Table 3.

TABLE 3 SOC VS TERMINAL VOLTAGE AT VARIOUS DISCHARGING CURRENT

S O C	At Discharging				
	Voltage at (C/150)	Voltage at (C/75)	Voltage at (C/50)	Voltage at (C/30)	Voltage at (C/15)
10	21.2	21.22	21.23	22.6	24
20	22	22.7	22.8	22.82	23.2
30	22.4	22.9	23	23.2	23.3
40	22.6	23.2	23.3	23.4	23.6
50	22.8	23.3	23.4	23.5	23.7
60	22.9	23.5	23.6	23.9	24.2
70	23	23.9	24.2	24.4	24.4
80	23.2	24.2	24.3	24.45	24.8
90	23.3	24.3	24.4	24.6	25
100	23.6	23.4	24.5	24.8	25.2

IV. HARDWARE IMPLEMENTATION

For the purpose of evaluating hybrid solar-wind system with the cuk-buck converter, a real time implementation of the conventional hybrid system wherein individual cuk and buck converters are connected to solar and wind system respectively is also implemented. Both the topologies are tested under the same seasonal conditions and input power. The input, output, battery voltage and current, energy flows between devices are measured to enable accurate assessment of the actual operating efficiencies. Since the hardware

responds in real time, a control algorithm is required to guide the operation of the system. So, FPGA based centralized control architecture is implemented. Control aspects related to maintain battery voltage, current, state of charge are also probed for proper energy management.

Research block roof top, SSN college of Engineering, kalavakkam in Chennai district in the Indian state of Tamilnadu is considered. The area around the place is partially sea shore and flat plain. The latitude and the longitude of the selected place are 12 ° 75’ and 80 °19’. The average weather of the particular place is 36°C, 60% humidity and the average wind velocity is 3m/s. So, solar and wind system can be placed in the selected site.

Fig. 10 and Fig. 11 shows the view of conventional converter based solar wind system and single stage cuk-buck fused converter setup. The 6 panels are connected in 3 parallel paths with 2 panels in series in each path. Based on the criteria, wind turbine of 1.4 kW, the rated wind speed of 15 m/sec, the cut in speed of 3 m/sec is considered. It has permanent magnet synchronous generator type with 48 V DC rectifier circuit. 4 numbers of 12 V batteries are connected in series to get 48 V. Load is connected with the point of common coupling through the battery bank.

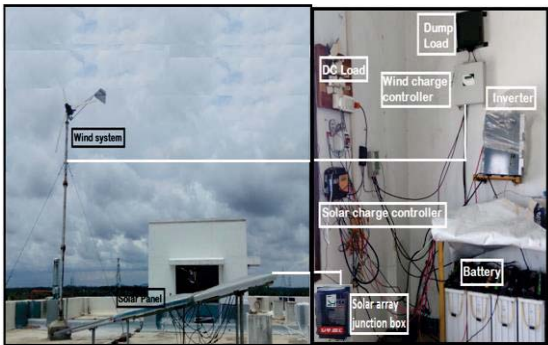


Fig. 10 Hardware setup for conventional system

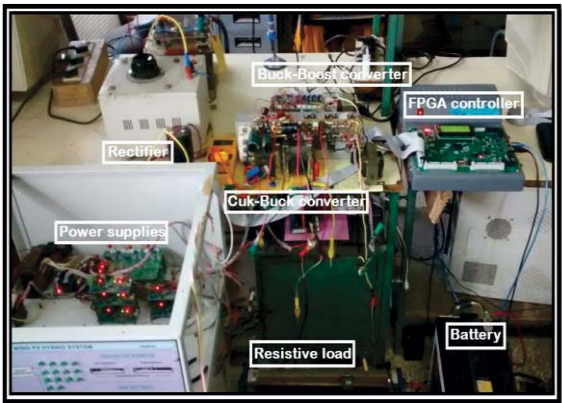


Fig. 11 Cuk-buck fused converter with FPGA controller

Specifications of the components used in the proposed work are given in Table 4.

TABLE 4 SPECIFICATIONS OF THE COMPONENTS

Specifications of components			
		Rating	Specifications
Inductors (Cuk-Buck)	L ₁	158μH	100 V, 100 A
	L ₂	2000 μH	100 V, 100 A
Capacitors	C ₁	2350μF	100 V, 100 A
	C ₂	2200μF	100 V, 100 A
Switch (Cuk-Buck)	S ₁	MTY100N10E	100 V, 100 A
	S ₂	MTY100N10E	100 V, 100 A
Switch (Buck)	S ₃	IRF250	100 V, 30 A
Switch (Boost)	S ₄	IRF250	100 V, 30 A
Switch- (Charging & Discharging)	S ₅	IRF250	100 V, 30 A
	S ₆	IRF250	100 V, 30 A
Load(Resistive)	DC	-48V,23A	1200 W
Input energy sources	V ₁	64V, 24 A	Solar energy
	V ₂	55V, 25A	Wind energy
Battery	V _B	12V-4 nos	200 Ah
Frequency		20kHz	

In the year 2016-17, the field study was performed and the findings are given in Table 5.

TABLE 5 FIELD FINDINGS (2016 MAY TO 2017 APRIL)

Field findings (2016 May to 2017 April)					
Month		P _s	Units	P _w	Units
		Watts	kWhr	Watts	kWhr
2016	May	461.70	69.26	42.20	4.22
	June	430.36	64.55	40.53	4.05
	July	466.47	69.97	48.96	4.90
	August	434.09	65.11	43.05	4.31
	September	385.67	57.85	101.49	10.15
	October	498.16	74.72	104.29	10.43
	November	243.69	36.55	108.31	10.83
	December	190.09	28.51	194.60	19.46
2017	January	194.75	29.21	42.17	4.22
	February	248.20	37.23	59.44	5.94
	March	504.26	75.64	59.83	5.98
	April	472.88	70.93	29.16	2.92

The solar, wind, battery, utility and inverter power obtained from the conventional system is given in the Fig 12. From the Fig it is found that the yearly average solar input power is 377.53 Watts and the yearly average wind input power is 72.83 Watts, the yearly average total power produced by hybrid energy system is 450.36 Watts and the yearly average inverter output power is 164 Watts. Yearly average battery power is calculated as 752 Watts.

The power at the point of common coupling is maintained constant by adjusting the duty cycles of the converter switches S₁ and S₂. Once input voltage is set for

solar and wind systems, the FPGA controller compares the load current with the solar current and wind current. The difference between the input currents and the load current is charging the battery through buck converter.

During the charging mode, the duty cycle of the switch in the buck converter is adjusted using the controller program. When load current is higher than the supply currents, battery is discharged along with the input currents through boost converter. The duty cycle of the switch in the boost converter is also adjusted using controller program. State of charge of the battery is also estimated using battery voltage and current. Controller is programmed to discharge the battery only if the state of charge of the battery is more than 40%. Fig. 13 shows the discharging current Vs state of charge of the battery for the year May 2016- April 2017.

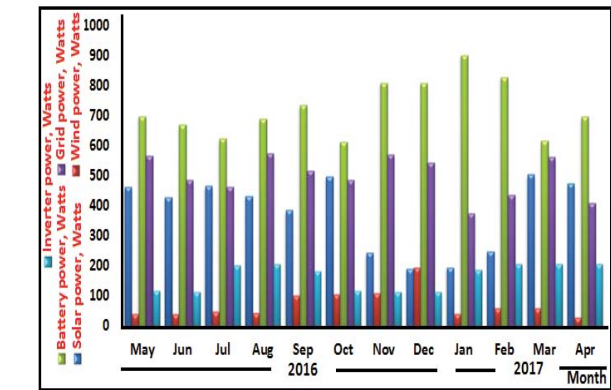


Fig. 12 Solar, wind, battery, utility, inverter power from conventional system

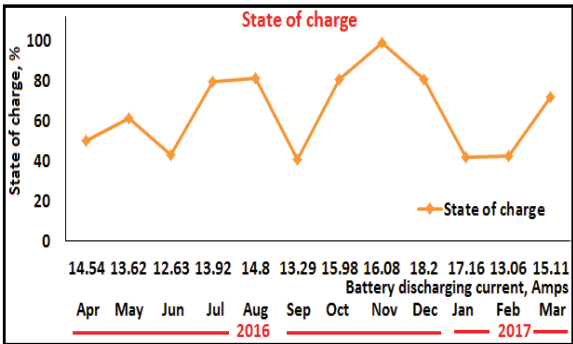


Fig. 13 Battery discharging current Vs Battery state of charge

Fig. 14 and Fig. 15 shows the solar, wind and battery input voltage and currents, output load voltage and current of cuk-buck converter. From the Figs, it is clear that the produced supply currents from hybrid renewable energy sources ($7.65\text{ A} + 1.79\text{ A} = 9.44\text{ A}$) are not able to meet the required load current. So, battery current ($I_b=16.5\text{A}$) supports the required load current.

A. Calculation of efficiency

The efficiency of conventional and cuk-buck converter is calculated as the ratio of the output power to the input power. So, solar/wind/battery system has been realized with cuk-buck converter (for an hour/day) from May 15th to May 31st, 2017. The days which produces same input power from both the converters are considered for the comparison of efficiency. All the component values are also kept same wherever possible, to maintain uniformity. Table 6 shows the comparison of efficiency.

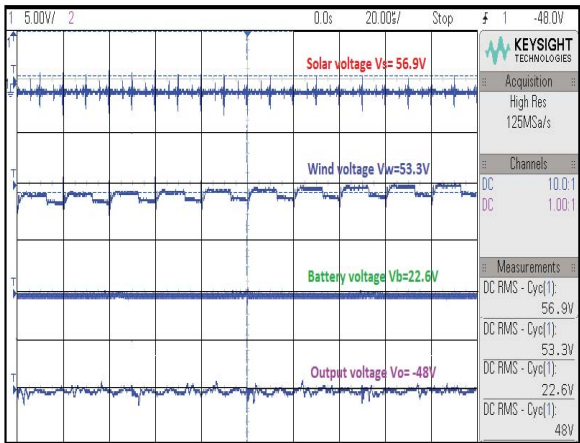


Fig.14 Input and Output load voltage

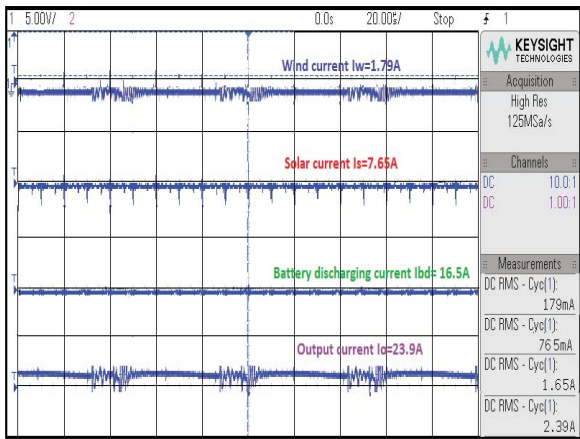


Fig. 15 Input and Output load current

TABLE 6. COMPARISON OF EFFICIENCY

Converter	Date	Sol ar	W in d	Total power		S O C	DC Load			
		P_s	P_w	$P_s + P_w$	P_b		$V_d c$	$I_d c$	$P_d c$	Eff
		W	W	W	W		V	A	W	$\%$
Conventional	10-09-2016	386	99	485	366	81	47	19	949	77
	05-12-2016	190	195	385	405	98	47	20	968	81
	19-01-2017	195	42	237	448	80	49	19	987	86
	27-03-2017	504	60	564	305	42	47	20	992	83
Cuk-buck	16-05-2017	346	42	388	383	43	48	20	1014	93
	18-05-2017	493	65	557	365	78	47	24	1135	91
	19-05-2017	429	57	486	350	93	47	22	1068	93
	26-05-2017	221	0	221	522	41	46	21	1021	95

As given in the Table, the total input (solar and wind) power on 10th September, 2016 and 19th May, 2016 are same. So, the efficiencies of the same is calculated and used for comparison. It is clear that the efficiency of cuk-buck converter (93.5%) is higher than the conventional converter (77.59%). In the same way other the efficiencies are also compared. Thus it has been proved that the efficiency of the dual input cuk-buck converter is the best one.

V. CONCLUSIONS

Thus the renewable energy interfaced single stage converter which provides uninterrupted supply with maximum efficiency, less cost, compact topology and less number of components. An embedded controller based on state of charge estimation, charging and discharging of batteries to suitably source or sink the input power according to the load demand is also presented. Experimental results based on solar system with 1.5kW peak power, wind system with 1.4 kW and 48V, 200 Ah Lead-Acid batteries validates the theoretical approach.

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